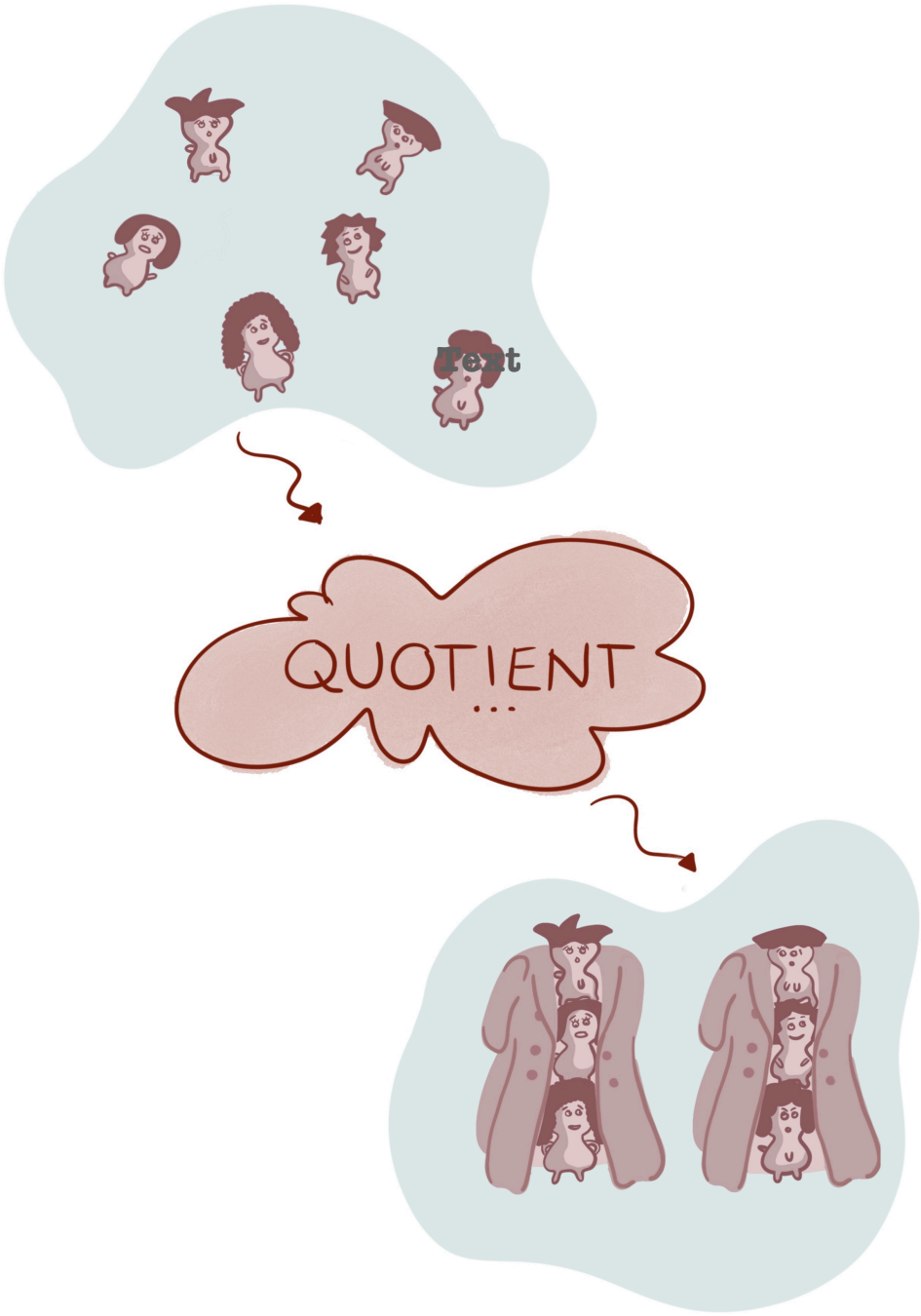
Definition Recall

Remember, last time we learned that a **normal subgroup** N of a group G is a subgroup such that, when you quotient G by N, you still get a group.



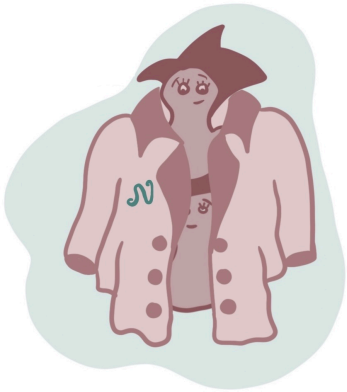
(Think about how *the divisors* of an integer are the numbers such that, after you divide the integer by a divisor, you still end up with an integer!)

Today's Group Theory Puzzle

If you take a group G....



...and you quotient it by a normal group N that is isomorphic to C_2 (the cyclic group of order 2)...



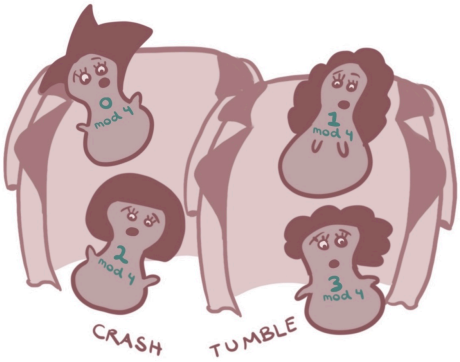
...and you get a group isomorphic to C_2...



...then can you reveal that the group G you started with...



...must be C_4?



Here is the full puzzle stated formally:

Consider a group G and a normal subgroup N.

If $G/N \simeq C_2$ and $N \simeq C_2$, then is $G \simeq C_4$?

Hint

Since $\frac{|G|}{|N|} = |C_2|$, then $\frac{|G|}{2} = 2$. So, $|G| = 4$.



We know G could be C_4.

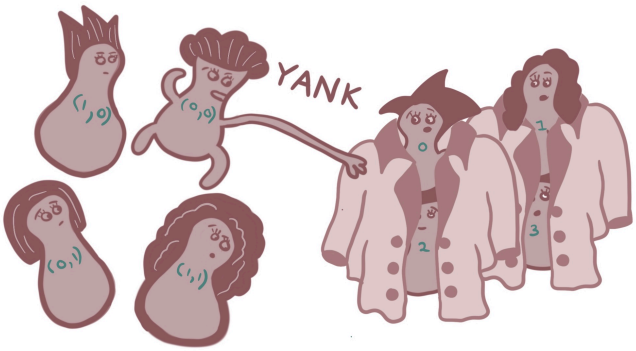


And the only other group of size 4 is $C_2 \times C_2$. Could that be G's true form?



Answer

The statement is *False* because G could also be $C_2 \times C_2$.



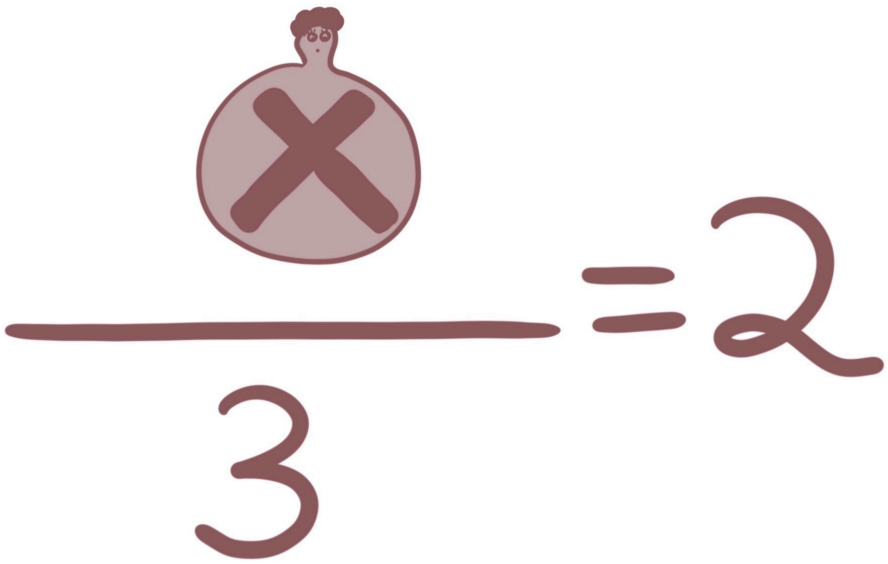
The quotient $|C_2 \times C_2|/|C_2|$ is isomorphic to C_2.



So: if $G/N \simeq C_2$ and $N \simeq C_2$, , then G could be $C_2 \times C_2$ or C_4.

Takeaway

With numbers, knowing the divisor and quotient is enough to identify the initial quantity.



But with group elements, it isn't so.

